Samarth Gupta

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Profile Summary

- Backend Engineer with **3 years** of hands-on experience building **scalable, high-performance backend systems** using **Python**.
- Strong background in **cloud-native architectures** using **AWS Lambda, EKS, Docker, and API Gateway**.
- Proven expertise in **low-latency system design** and building **efficient, scalable data pipelines**.
- Demonstrated ownership in optimizing **performance, cost, and reliability** in fast-paced product environments.
- Experienced in designing and owning **microservices, REST APIs, and real-time decisioning platforms** across **serverless and containerized environments**.

Experience

Backend Engineer September 2024 – Present
Fibe.India

- **Blackbox Decisioning Platform**
  - Built real-time decisioning microservices delivering **approve/reject** outcomes using rule engines and scorecard-based logic.
  - Designed **dual compliant deployments**: obfuscated on-prem Docker images and serverless **AWS Lambda + API Gateway** architecture.
  - Secured critical business logic through **100% Python obfuscation (PyArmor)** and distributed Docker **.tar** images to prevent IP exposure.
  - Automated testing of Docker images across thousands of customer scenarios, boosting validation efficiency by **90**% and ensuring reliable, error-free partner deployments.
  - Achieved **zero-downtime autoscaling** using Lambda and API Gateway, handling traffic spikes with automatic failover.
  - Reduced manual deployment and operational overhead by **80%** through automated build, obfuscation, and packaging pipelines.
  - Improved external partner onboarding time by **60–70%** using standardized API contracts and integration patterns.

- **Real-Time ML & Large-Scale Serving Architecture**
  - Architected high-performance ML inference services for **7–8 GB models** using Lambda-based orchestration with autoscaled **EKS pods**.
  - Reduced end-to-end latency from **2 minutes to ¡1 second** by decoupling orchestration (Lambda) from heavy computation (EKS).
  - Deployed credit and debit model services on a shared EKS cluster with **HPA-enabled pods** for independent horizontal scaling.
  - Exposed inference services via **NLB/ELB**, enabling efficient traffic distribution across multiple pods.
  - Implemented **async parallel execution** in Lambda using **asyncio + aiohttp**, aggregating results from **4 concurrent endpoints**.
  - Optimized model initialization using **Joblib** and pod-level loading, eliminating repeated model load overhead.
  - Built vectorized **Pandas/NumPy** pipelines, reducing data processing time by **60%** and achieving **¡800 ms** total latency.
  - Enabled secure configuration via **AWS Secrets Manager** and real-time monitoring using **CloudWatch Logs → Subscription Filters → Grafana**.
  - Delivered a cost-efficient architecture combining **pay-per-request Lambda** with **autoscaled EKS** for high availability and fault isolation.

Software Development Engineer May 2023 – August 2024
Consultadd Inc.

- Leveraged **AWS ECS** to orchestrate and manage a fleet of microservices, achieving **30%** reduction in operational costs through efficient resource utilization.
- Implemented **AWS Systems Manager Parameter Store** to securely manage environment-specific configuration, reducing deployment time by **20%** and improving application reliability.
- Worked on processing messages from **Amazon Simple Queue Service (SQS)** subscribed to **Amazon Simple Notification Service (SNS)**, efficiently handling message consumption and performing necessary processing based on message content.
- Leveraged **SQL and MongoDB** to efficiently retrieve data and transforming it into structured CSV format using Python scripting and to seamlessly upload the processed CSV files to Amazon **S3** and **SFTP** servers.
- Developed and optimized **MNT** file generation scripts to process over **1 million** records daily. Implemented error handling and validation to ensure data accuracy. Automated MNT file loading into Xstore, reducing manual intervention by **80%** and improving data update time by **35%**.
- Developed and implemented new features and enhancements for web applications using Python, Django.
- Optimized **database queries** and system performance to improve overall application speed and user experience.
- Worked in an agile environment using **JIRA** and documented all the new processes and tasks using **Confluence**.

Software Development Engineer Intern January 2023 – April 2023
Consultadd Inc.

- Developed API to perform CRUD operation between database and application.
- Utilized **Python, Django, AWS, SQL, Git, and GitHub** to craft and execute software solutions, enhancing system efficiency and performance.
- Remained at the forefront of technology by researching and seamlessly integrating new tools and technologies, with a focus on optimizing software development processes and leveraging AWS services for project success.

Technical Skills

Languages: Python, SQL
Backend & APIs: Flask, Django, RESTful APIs
Cloud & Infrastructure: AWS (Lambda, EKS, ECS, EC2, SQS, SNS, API Gateway, CloudWatch, SSM, S3), Docker
Databases: MySQL, MongoDB, IBM DB2, RDS, DynamoDB
Tools: Git, JIRA, Confluence

Certification

AWS Certified Solutions Architect Associate —Link June 2023

Education

Acropolis Institute of Technology and Research Indore, India
B-Tech in Computer Science 2019 - 2023
CGPA: 8.39

Achievements

1. Honored with the **Pinnacle Award** for positive contributions during the quarter.
2. Won **Employee of the Month** title for three times.
3. Won **Star Performer of the Month** title for three times.